Mithun Kumar J

Embedded Systems · Digital Hardware · RF Engineering

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PROFILE

Founder of SYNDRIX and third-year Electronics and Communication Engineering undergraduate at VIT Chennai. Hands-on work across embedded control, IoT telemetry, FPGA receiver decoding, wearable PCB design, and antenna simulation.

EDUCATION

VIT Chennai — B.Tech, Electronics and Communication Engineering

2024–2028 · Expected graduation 2028

Focus: Embedded Systems, Digital Hardware Design, FPGA, VLSI, RF Engineering, and Edge AI.

EXPERIENCE & LEADERSHIP

Founder — SYNDRIX

APR 2026–PRESENT

Electronics and embedded-systems venture focused on PCB design, firmware, IoT solutions, prototyping, and software integration.

Embedded Systems & IoT Intern — AKVO ATMOSPHERIC WATER SYSTEMS

2026

Raspberry Pi-based AWG control, sensor integration, finite-state-machine automation, FastAPI, Supabase cloud telemetry, and fault handling.

Electronics Team Member — SHAURYA RACING · VIT CHENNAI

FORMULA EV

Formula EV electronics, ESP32-based CAN telemetry, sensor interfaces, and hardware integration.

Member — IEEE ROBOTICS & AUTOMATION SOCIETY

VIT CHENNAI

Part of the robotics and automation community at VIT Chennai.

RECOGNITION

- Smart India Hackathon 2026 — Internal Round Top 50 among approximately 1,700 participating teams at VIT Chennai.
- IIT Madras Shaastra Robo Soccer opening-round winner and finalist, Team GOAL GORITHM
- Participated in six or more hackathons

SELECTED ENGINEERING PROJECTS

SMART GLASSES / RF ENGINEERING

SIMULATED & EVALUATED · RF engineering / Antenna simulation / 2.45 GHz

A compact 2.45 GHz planar dipole antenna designed and simulated for smart glasses. Evaluated impedance matching, radiation performance, and human-head interaction.

−18.8 dB S11 · 45.5% radiation efficiency in on-head simulation

LDACS / FPGA RECEIVER

IMPLEMENTED & VERIFIED · FPGA / Vivado / Zynq-7000

Implemented and verified an LDACS receiver decoding architecture on Xilinx Vivado and Zynq-7000, including channel estimation, equalization, and PS/PL integration.

9 verification cases completed · Timing met

AKVO / ATMOSPHERIC WATER SYSTEMS

DEVELOPED · Python / FastAPI / Supabase / Raspberry Pi

Raspberry Pi/Linux control software integrating temperature, humidity, water-flow, and tank-level sensors. Finite-state-machine automation, fault handling, and cloud telemetry.

Sensors → State machine → Actuators → Cloud

WEARABLE / FOUR-LAYER HARDWARE

DESIGNED · KiCad / STM32 / PCB design

A four-layer STM32 wearable PCB in KiCad, bringing GPS, power distribution, and RF-aware routing into a compact hardware design.

4-layer PCB · STM32 · GPS · RF-aware routing

ONGOING DEVELOPMENT

RTL Verification & AI Accelerator

Learning Verilog RTL, sequential logic, simulation, testbench development, and hardware verification. Working toward an INT8 MAC accelerator and eventual ASIC-oriented flow.

ADDITIONAL PROJECTS

ESP32 Antenna — antenna design and simulation in CST Studio Suite; Robo Soccer Striker — ESP32, ESP-NOW, motor drivers, custom wireless drive controls; DECAD — Schematic OCR and circuit reconstruction; RFID Attendance System — ESP32, RC522, Python, Google Sheets; FineHands — Financial document analysis using OCR, ML, and Supabase; Ship Spy Live — YOLO-based vessel detection and monitoring; Delta Modulation FPGA — ZedBoard encoder and decoder.

Updated November 2026. Antenna results are on-head simulations. Ongoing work is listed separately from completed implementations.